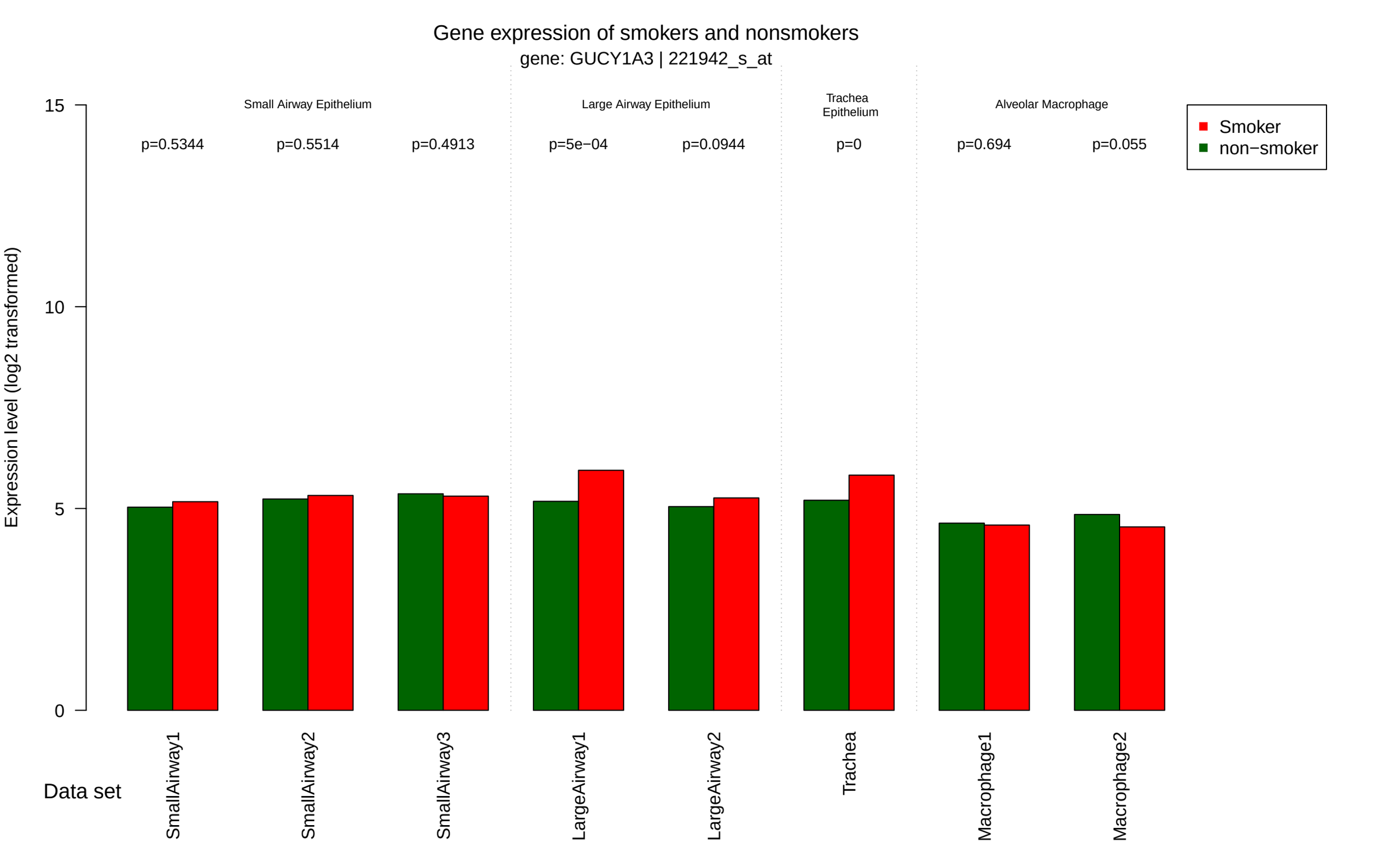
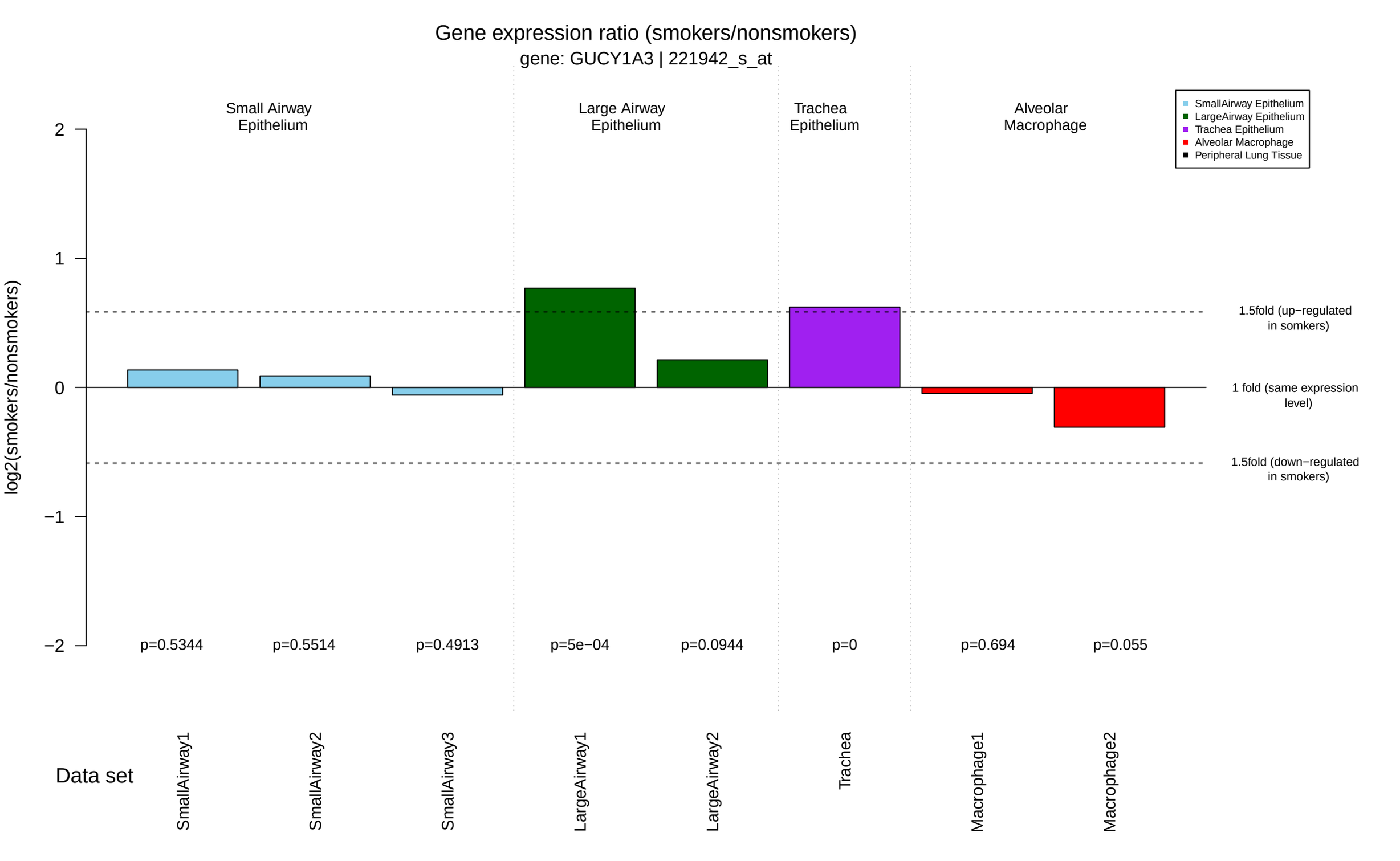
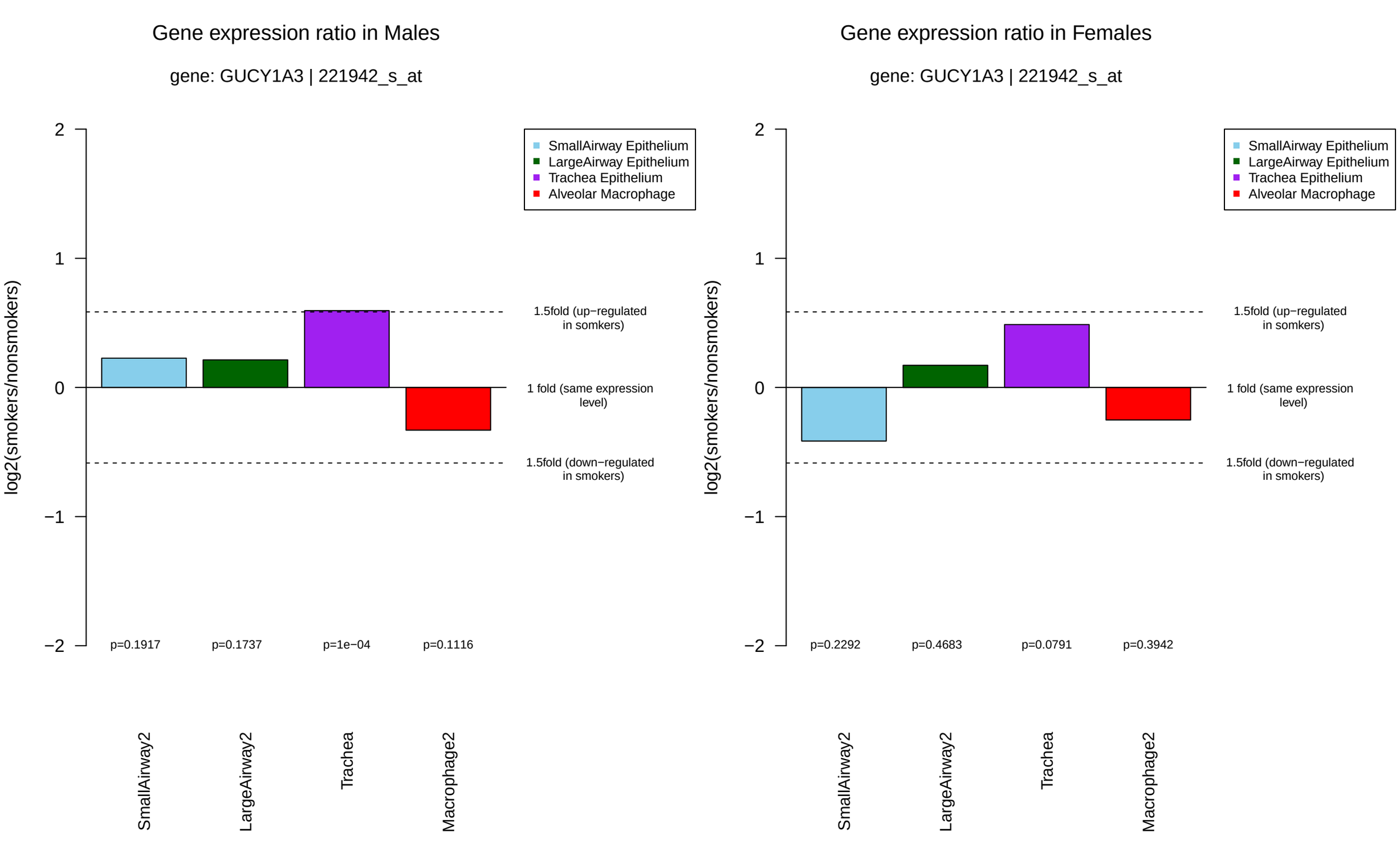
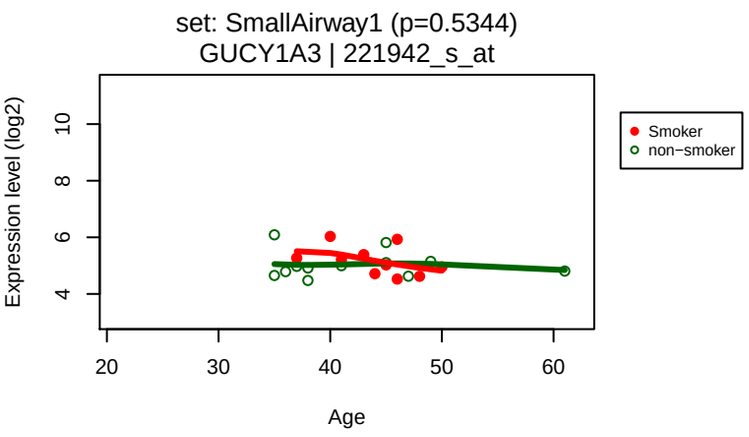
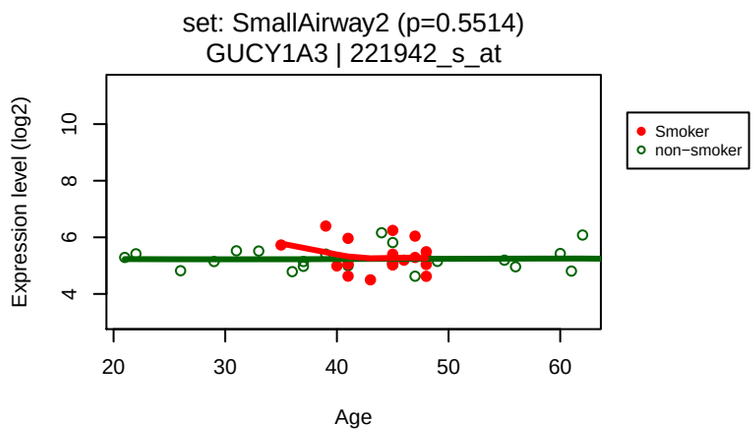
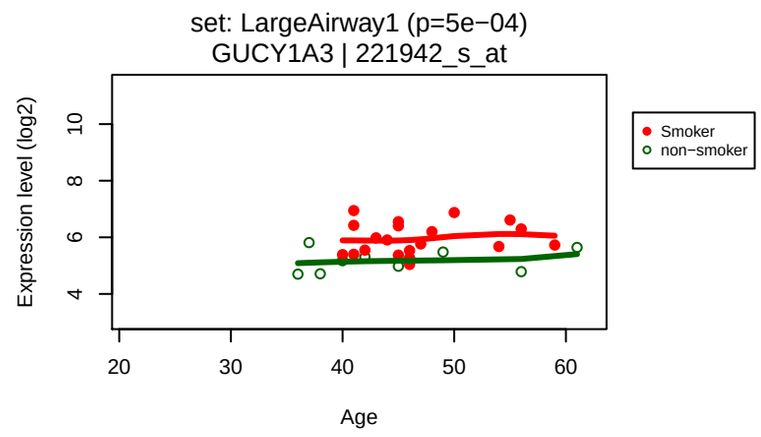

Gene expression of smokers and nonsmokers
gene: GUCY1A3 | 221942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5344
p=0.5514
p=0.4913
p=5e−04
p=0.0944
p=0
p=0.694
p=0.055
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

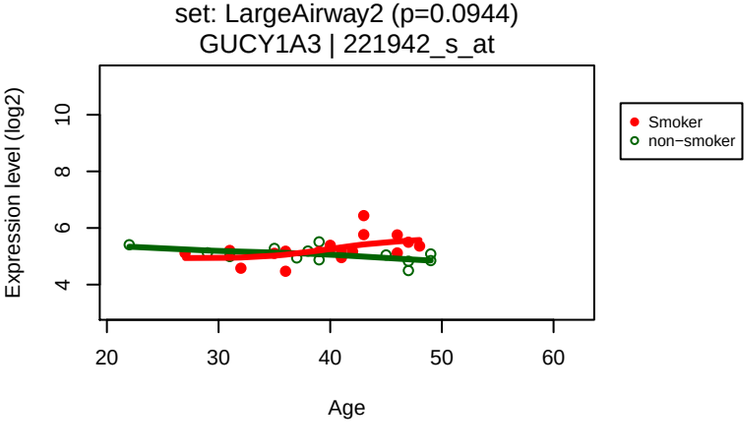

Gene expression ratio (smokers/nonsmokers)
gene: GUCY1A3 | 221942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5344
p=0.5514
p=0.4913
p=5e−04
p=0.0944
p=0
p=0.694
p=0.055
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

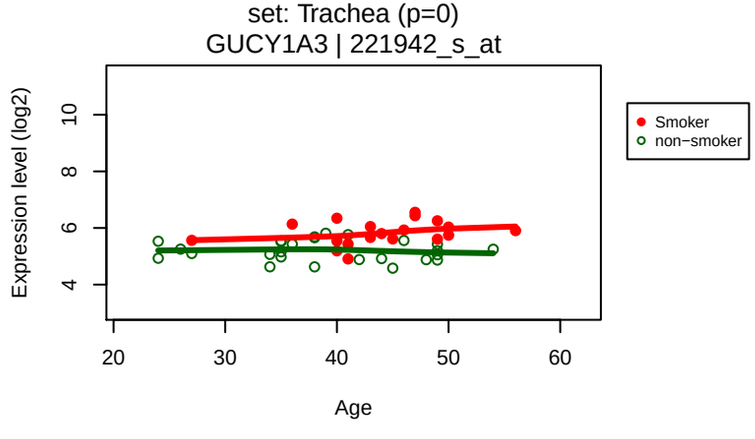

Gene expression ratio in Males
gene: GUCY1A3 | 221942_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1917
p=0.1737
p=1e−04
p=0.1116
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GUCY1A3 | 221942_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2292
p=0.4683
p=0.0791
p=0.3942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

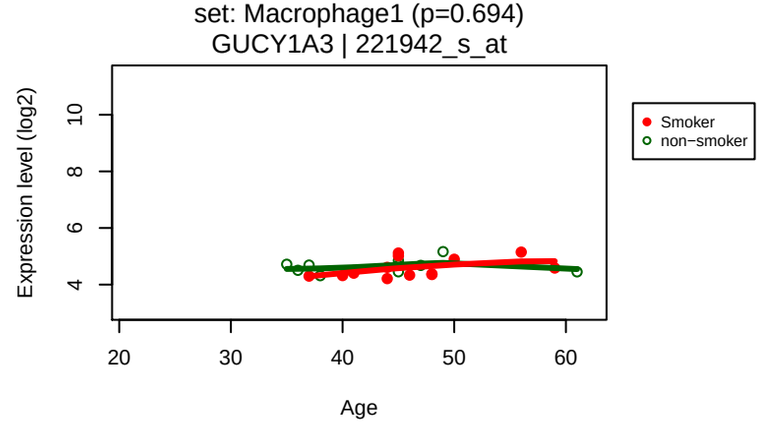

set: SmallAirway1 (p=0.5344)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker

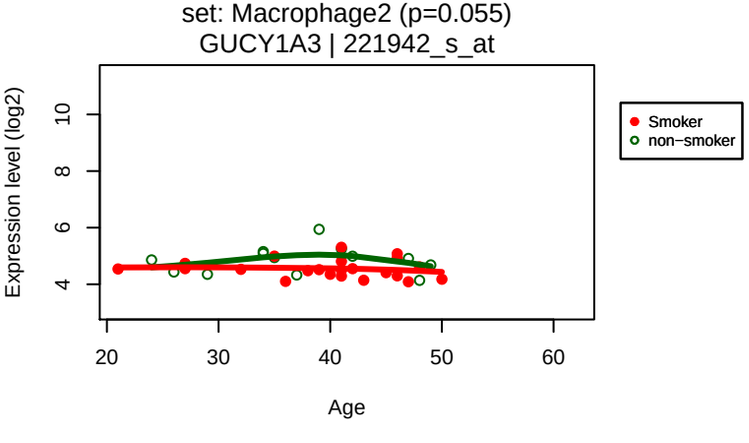

set: SmallAirway2 (p=0.5514)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker

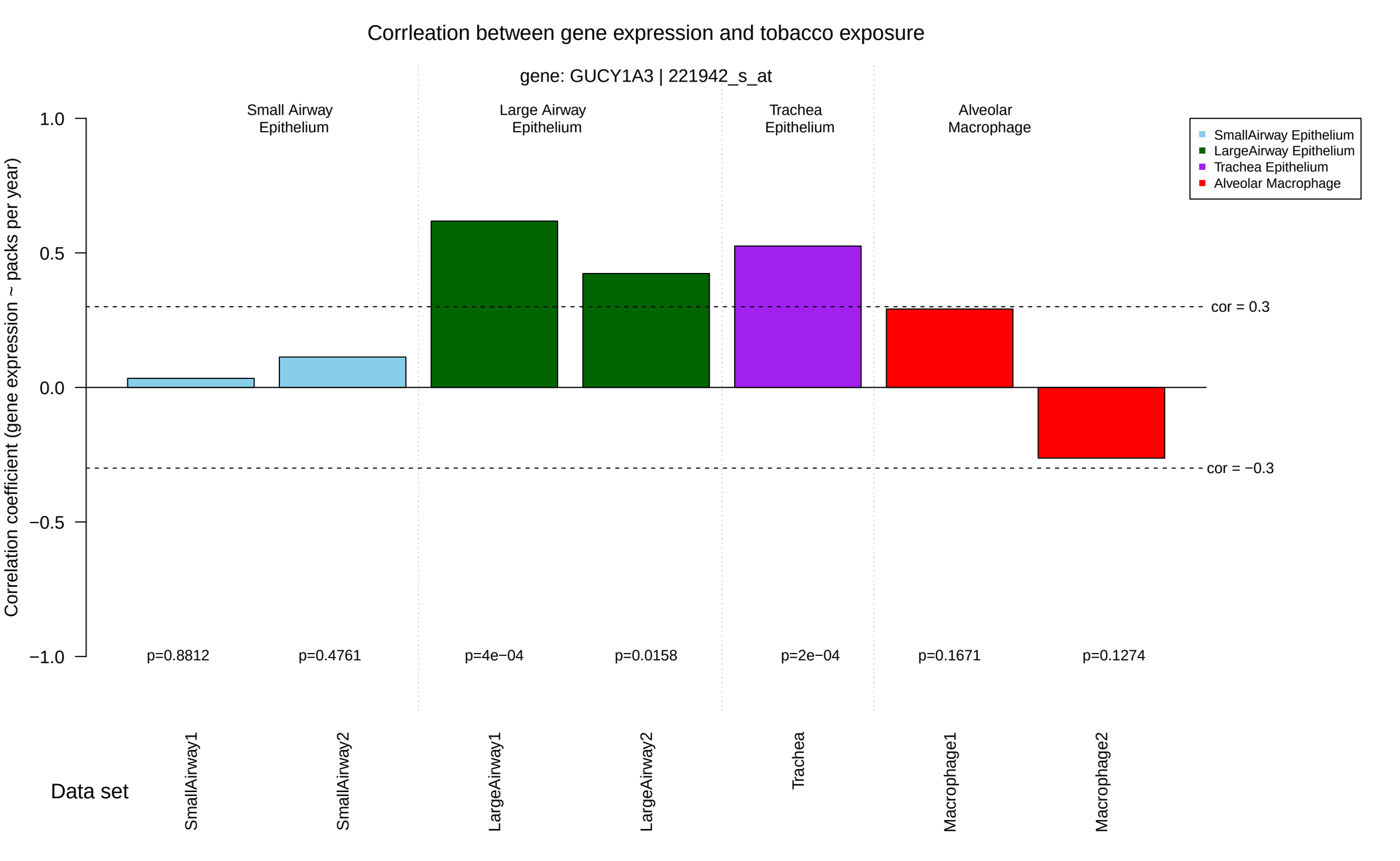

set: LargeAirway1 (p=5e−04)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0944)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.694)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.055)
GUCY1A3 | 221942_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GUCY1A3 | 221942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8812
p=0.4761
p=4e−04
p=0.0158
p=2e−04
p=0.1671
p=0.1274
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2